Exercise 1

It is known the directivity function of an antenna operating at 800 MHz:

$f(\theta)=\frac{3\cdot\sqrt{3}}{2}\sin(\theta)\cos^2(\theta)$ for $0<\theta<90°$, $f(\theta)=0$ elsewhere.

1) Compute the value of θ where f is maximum (θ_{max}).
2) Compute the directivity gain D_M. Recall: $\int\cos^2(x)\cdot\sin^2(x)dx=\frac{x}{8}-\frac{\sin(4x)}{32}$
3) Evaluate the minimum efficiency of the antenna which determines the gain G equal to 5 dB
4) Assuming the antenna used in transmission with P_T=100W, evaluate the distance from the antenna (along the direction of maximum emission) where the radiated power density S_R is 10^{-3} W/m^2. What is the electric field intensity (V/m) at this distance?

Solution:

1) Derivative of f(θ) equal to zero:

$$\frac{df}{d\theta}=\cos(\theta)-3\sin^2(\theta)\cos(\theta)=0 \quad\Rightarrow\quad \theta_{\max}=\sin^{-1}\left(\sqrt{\frac{1}{3}}\right)=0.6155 \text{ rad}$$

2) D_M computation:

$$D_M=\frac{4\pi}{\frac{3\sqrt{3}}{2}\int_0^{2\pi}\int_0^{\pi/2}\sin^2(\theta)\cos^2(\theta)\,d\varphi d\theta}=\frac{4}{3\sqrt{3}}\frac{1}{\left[\frac{\theta}{8}-\frac{\sin(4\theta)}{32}\right]_0^{\pi/2}}=$$

$$=\frac{4}{3\sqrt{3}}\frac{16}{\pi}=3.9 \quad (5.93 \text{ dB})$$

3) $\eta=\frac{G}{D_M}=0.806$

4)
$$S_R=\frac{P_T G}{4\pi R^2}\Rightarrow R=\sqrt{\frac{100\cdot10^{0.5}}{4\pi10^{-3}}}=158.63 \text{ m}$$

$$E=\frac{1}{R}\sqrt{\frac{Z_0\cdot P_T\cdot G}{2\pi}}=\frac{1}{158.63}\sqrt{\frac{377\cdot100\cdot10^{0.5}}{2\pi}}=0.87 \text{ V/m}$$

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$f = 800\,MHz$, $f(\theta,\varphi) = \frac{3\sqrt{3}}{2}\sin\theta\cos^2\theta$ $\quad \theta \le 90°$ (θ downwards)

① Find θ $f_{MAXIMUM}$:



$$\frac{\partial f(\theta,\varphi)}{\partial\theta} = \frac{3\sqrt{3}}{2}\left(\cos^3\theta - 2\sin^2\theta\cos\theta\right)$$

$$= \frac{3\sqrt{3}}{2}\cos\theta\left(\cos^2\theta - 2\sin^2\theta\right)$$

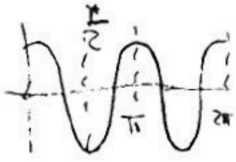
$$\Rightarrow \frac{\partial f}{\partial\theta} = 0 \Rightarrow \begin{cases} \theta = 90° \\ \vee \\ \cos^2\theta = 2\sin^2\theta \Rightarrow \tan^2\theta = \frac{1}{2} \Rightarrow \tan\theta = \frac{1}{\sqrt{2}} \Rightarrow \end{cases} \begin{cases} \theta_{max} = 0.615\,rad \\ = 35.2° \\ f(\theta,\varphi) = 1 \end{cases}$$

② D_{MAX}: $\quad S_R(r,\theta,\varphi) = \frac{P_{RAD}}{4\pi r^2} D_R \cdot f(\theta,\varphi)$

$$\Rightarrow D_R = \frac{4\pi}{\int_0^{2\pi}\int_0^{\pi} f(\theta,\varphi)\sin\theta\,d\theta\,d\varphi} = \frac{4\pi}{2\pi\int_0^{\frac{\pi}{2}} \sin^3\theta\cos^2\theta\,d\theta}$$

$$\int \sin^2\cos^2 = \int \sin^2 - \int \sin^4 = \int\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) - \int\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)^2$$

$$= \int\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) - \int\left(\frac{1}{4} + \frac{1}{4}\cos^2 2x - \frac{1}{2}\cos 2x\right)$$



$$= \int\frac{1}{2} - \int\frac{1}{4} - \frac{1}{4}\int\cos^2(2x) = \int\frac{1}{2} - \int\frac{1}{4} - \frac{1}{4}\int\left(\frac{1}{2} + \frac{1}{2}\cos 4x\right) = + \int\frac{1}{8}dx$$

$$\Rightarrow D_R = \frac{4\pi}{2\pi\,\frac{1}{8}\frac{\pi}{2}\cdot\frac{3\sqrt{3}}{2}} = 3.9\ (= 5.9\,dB)$$

③ MIN EFF for $G = 5\,dB$

$$G = \eta \cdot D \qquad \eta = \frac{G}{D} = 0.813$$

④ $P_t = 100\,W$: find R where $S = 10^{-3}\,\frac{W}{m^2} = 1\,\frac{mW}{m^2}$ and compute $|E|$ (in direction of MAX PROPAGATION)

$$S = \frac{P_t \cdot D_{max}\,\eta}{4\pi R^2} \Rightarrow \qquad R = \sqrt{\frac{P_T D_{max}\,\eta}{4\pi S}} = 1.5\,m$$

$$S = 1\,\frac{mW}{m^2} = \frac{1}{2}\,\frac{|\bar{E}|^2}{\varepsilon_0} \Rightarrow |\bar{E}| = \sqrt{2\varepsilon_0 \cdot 1\,\frac{mW}{m^2}} = 0.86\,\frac{V}{m}$$

RECALL $\bar{S}$ and $\bar{E}$:

$$\bar{S} = \frac{1}{2}\,\bar{E} \times \bar{H}^* \qquad \bar{H} = \frac{\hat{e}_z \bar{E}}{Z_0} \Rightarrow S = \frac{1}{2}\,\frac{|\bar{E}|^2}{Z_0}$$

$\bar{E}$: PHASOR of electric field $\qquad \eta = 377\,\Omega$ in AIR, VACUUM

$$S = \frac{P_T\ \text{flow}}{4\pi R^2} D_{max} \cdot \eta = \frac{P_T}{4\pi R^2} D_{max}\,\eta \Rightarrow R = \sqrt{\frac{P_T D_{max}\,\eta}{4\pi S}}$$